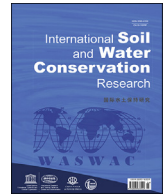




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Original Research Article

Influence of post-fire root decay-induced soil macropores on slope stability: A new method for analyzing heterogeneous slope stability

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ABSTRACT

Soil macropores are key factors affecting slope hydrological processes and stability, particularly under heavy rainfall conditions. Although wildfires can lead to the decay and death of plant roots, leaving root channels, few studies have examined temporal variation in the distribution of soil macropores or their impact on slope stability. To address this, we examined the bacterial abundance, root distribution, and macropore characteristics of burnt forest at one week and 6 and 12 months post-fire. Numerical simulation was used to analyze the effects of macropore distribution on slope stability under extreme rainfall conditions (80 mm/d × 4 d) at each time-point. Soil macropores accelerated the propagation of water pressure, potentially triggering shallow-slope instability. In the simulation, following 1 d of rainfall, slope stability was lower, by 3.55% and 8.68%, respectively, at 6 and 12 months than at one week post-fire. Following 4 d of rainfall, slope stability was better at 6 and 12 months than at one week post-fire, by 1.87% and 2.81%, respectively, owing to the drainage effect of the macropores. Even more importantly, this study proposed a method for coupling the spatial heterogeneity of soil macropores with a numerical model of slope stability. These findings help to elucidate the temporal changes in vegetated slope hydrology and stability after a wildfire and provide a reference for the numerical simulation of the stability of heterogeneous slopes.

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1. Introduction

Wildfires often result from surface-layer desiccation and heat accumulation during summer and autumn. Intense fires substantially reduce vegetation cover, thus reducing biodiversity and ecosystem functioning and potentially increasing the risk of surface instability and human activity in the burnt area (Bardgett et al., 2021). Post-fire decay of plant roots leads to a reduction in soil cohesion, and the root channels remaining after decomposition allow water to more easily infiltrate into deeper layers during

rainfall events (Chen et al., 2018). These factors render post-fire areas highly sensitive to secondary disasters such as shallow landslides and debris flows (Cui et al., 2021). The probability of landslides and debris flows occurring at post-fire sites was estimated to be as high as 40% (Parise & Cannon, 2012). For example, in 2018, a post-fire mudslide in California (USA) killed 23 people, injured at least 167 people, and damaged 408 houses (Kean et al., 2019). In 2019, a serious forest fire in Liangshan Prefecture (Sichuan Province, China), over an area of ca. 20 ha, triggered a mudslide that caused 31 deaths (Lei et al., 2022). Slope stability is controlled primarily by mechanical properties and hydraulic conditions (Dai et al., 2022; Pradhan et al., 2019). Previous studies on post-fire landslide triggering mechanisms have mostly focused on the degradation of root reinforcement, which leads to changes in soil shear strength (Abdollahi et al., 2023; Lei et al., 2022; Li et al., 2023). However, few studies have also addressed the post-fire

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temporal changes in soil hydrological behavior. Macropores created by root decay accelerate the propagation of pore water pressure, which is directly related to slope stability (Lu et al., 2022). Therefore, studying the distribution of macropores in post-fire forest soils and the associated impacts on the hydrological properties and stability of slopes is of significant importance.

Over both short and long timescales, wildfires affect the vegetation cover, soil properties, microbial activity, and subsurface storage (Zhao et al., 2024). Immediately after a fire, the soil macropores become blocked by the burnt plant matter and litter, which creates a hydrophobic layer on the ground, and the resulting reductions in water retention and hydraulic conductivity increase the likelihood of surface runoff (Moody et al., 2013). Plant roots are key factors in the effective regulation of soil structural stability and surface hydrological processes (Dai et al., 2022; Zhu et al., 2022). Intense burning reduces microorganismal abundance, and the decoupling of organic matter blocks the pathways whereby roots acquire resources, thus substantially reducing root activity (Montagnoli et al., 2023; Philippot et al., 2024). After a fire, rainfall washes away the hydrophobic layer on the surface and microbial activity gradually recovers. Microbial metabolic processes accelerate root decomposition and decay (Vergani et al., 2016, 2017). The removal of the hydrophobic layer by rainfall also causes the infiltration capacity of the burnt soil to increase as macropores develop (Liu et al., 2021). Owing to the reduction in runoff and increase in infiltration, shallow landslides are the primary mode of surface degradation in the years following wildfires (Parise & Cannon, 2012). The preferential channels formed by macropores can enhance vertical infiltration, accelerate the response of groundwater to rainfall, increase pore water pressure (PWP) on the potential slip surface, and reduce the effective shear strength (Shao et al., 2018). However, root decay and regrowth after wildfires alter soil macropore distribution and conductivity (Jiang et al., 2018). Nonetheless, the understanding of the spatiotemporal distribution of post-fire macropore formation and the impact of macropores on slope seepage fields remains incomplete.

Traditional landslide early warning systems often overlook the impact of macropores and preferential flows on slope stability, leading to underestimation of landslide disaster risks under heavy rainfall conditions (Shao et al., 2015, 2016). Unsaturated infiltration processes and the effects of preferential flow on slope-failure modes have become key focus areas and challenges in landslide early warning and monitoring systems (Moradi et al., 2018; Tao et al., 2020; Vivaldi et al., 2024). Modified permeability models, dual permeability models, and empirical equations describing the interaction between porosity and soil hydraulic conductivity have been widely developed and utilized in recent years (Gackiewicz et al., 2022; Shao et al., 2018; Yeh et al., 2021). Modified permeability models typically introduce an effective hydraulic conductivity to reflect uneven water flow paths (Dou et al., 2014), however, due to the complexity of the soil structure, the calculation of the parameters involves significant errors. Dual permeability models assume that water flow occurs independently in the macropore and matrix domains (Shao et al., 2015, 2018). While this approach provided effective simulations of the basic effects of preferential flow, it often involved linear assumptions, thereby overlooking the nonlinear interactions between soil and water. Furthermore, many existing studies assumed constant water exchange coefficients between the two domains, which is not applicable to soils under complex conditions (Sidle & Bogaard, 2016). More importantly, the distribution of macropores in soils is heterogeneous and random, and this method neglects the mixing effects of water flow paths, making it difficult to quantify the high heterogeneity of the soil (Jia et al., 2024). Although some researchers have observed the microstructure of soils using X-ray tomography and derived

nonlinear relationships between porosity and saturated hydraulic conductivity (Gackiewicz et al., 2022; Jiang et al., 2018; Yang et al., 2021), the study scales are small, and effectively observing the macropore flow patterns influenced by complex soil behaviors, such as root growth and decay, microbial activity, and organic matter changes, is challenging. Image analysis and dye tracing have been widely applied in the field to identify macropore flow paths in soil profiles (Ronchetti et al., 2020; Hou et al., 2023). In this study, numerical analysis was performed on images obtained from dye tracer tests to identify the macropore and matrix domains. A functional relationship between macropore aperture and saturated permeability was established based on the differences in macropore size. The impact of soil macropore effects on slope hydrological behavior was quantified.

This study aimed to quantify the impact of post-fire changes in soil macropores on the hydrological processes of slopes under extreme rainfall, via field experiments and numerical analysis. Continuous field investigations and experiments were conducted over one year on slopes affected by forest fires. We analyzed the impact of microbial community activity on root death and root-channel formation. Images of macropore seepage pathways were obtained via dye tracer testing. The dye images were expanded to the scale of the slope surface. The spatial distribution of macropore flow was analyzed numerically using finite element software. The hydrological processes of heterogeneous slopes were simulated using numerical methods that couple macropore flow pathways with hydrological mechanics models, and the point factor of safety method was employed to estimate slope stability at one week, 6 months, and 12 months post-fire. These findings provide a basis for early detection and warning systems for shallow landslides following forest fires.

2. Materials and methods

2.1. Study area and sites selection

The study area was located on Mount Jinyun in Beibei District, Chongqing, China (106°17′–106°24′ E, 29°41′–29°52′ N; elevation, 200–952 m). The region features a subtropical monsoon humid climate, with an average annual temperature of 13.6 °C and average annual precipitation of 1611.8 mm. Most of the rainfall (77.2% of total annual precipitation) occurs between April and September. Drought occurs from July to August, with August being the hottest month, with an average temperature of 24.3 °C. There is substantial evaporation in July and August, accounting for 32.8% of the total annual evaporation. The predominant soil type in the study area is sandy loam formed mainly by the weathering of sandstone. In 2022, Chongqing experienced high temperatures and little rain from July to August, and a major forest fire occurred in Hutou Village, Beibei District, from August 21 to 26, 2022, causing extensive forest damage. Prior to the wildfire, the predominant woodland type in the burned area was a secondary forest with *Pinus massoniana* Lamb. as the dominant species. Experimental sites were selected in severely burned areas (Fig. 1). Investigations and experiments were conducted one week, 6 months, and 12 months after the wildfire.

2.2. Blue dye tracing experiments and sample collection

The wildfire had a significant impact on plant and soil characteristics within 6 months–12 months after the event (Köster et al., 2021; Warneke et al., 2023). Three blue dye tracing experiments were conducted, on September 3, 2022, March 4, 2023, and September 3, 2023; no rain fell on those days or on the days immediately before or after the experiments (Fig. 2). The

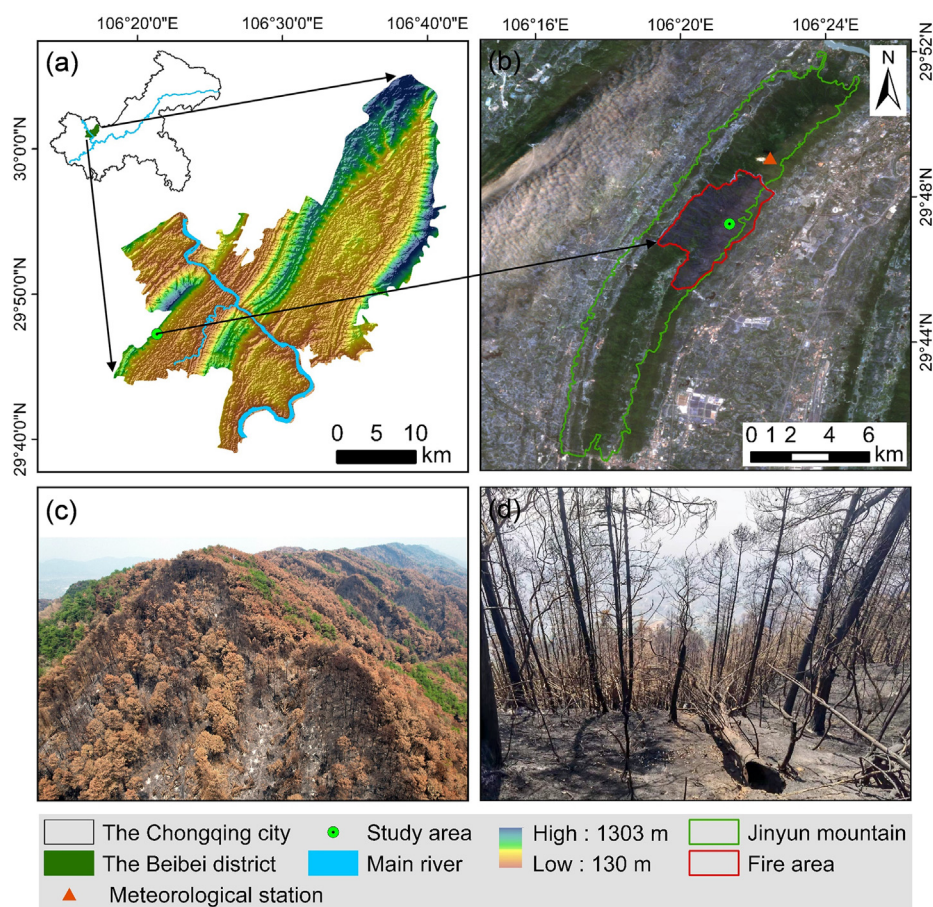


Fig. 1. Overview of the study area of the Jinyun Mountain in Hutou Village, Beibei District, Chongqing, China: (a) location of the study area; (b) experimental sites; (c) aerial photo of the burned area; (d) photo of the severely burned area, pines were burned and ash deposition covered the slope.

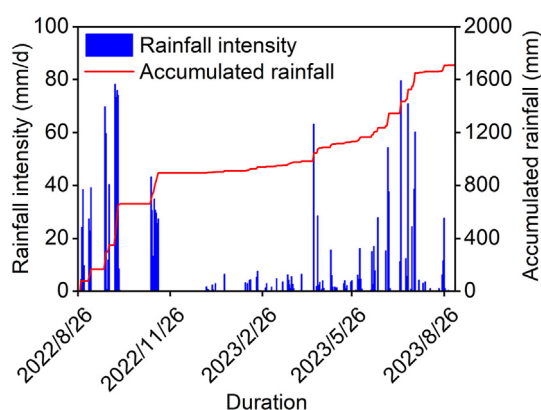


Fig. 2. Rainfall conditions during the year after the fire (data from Jinyun Mountain Meteorological Station).

experimental site was located 1 m laterally from a single *Pinus massoniana* (Fig. 3(a)). The surface layer of dead leaves and ash was removed, and the surface was leveled before the experiments commenced. An iron frame ($1.1 \times 1.1 \times 0.3$ m) was inserted vertically into the soil to a depth of 0.2 m, and the soil within 0.05 m of the inner wall of the frame was compacted to ensure that the water infiltrated vertically through the soil. Brilliant Blue FCF (4 g/L) was used to prepare 40 L of solution to be sprayed uniformly on a 1×1 m area of soil surface inside the iron frame

was covered with a plastic film to minimize the effects of evaporation and natural rainfall after the solution was sprayed. Soil observation profiles were excavated along the ground at a distance of 1 m from the trees after 24 h of infiltration (Hencher, 2010; Zhang et al., 2017), parallel to the dip of the slope. The area within each profile where the dye was considered the stained area, whereas the area with little or no dye was considered the blank area (Fig. 3(b)) (Zhang et al., 2017; Adobe Photoshop 2023 (Adobe, San Jose, CA) was used to pre-process the captured images to correct the illumination and chromatic aberration. The grayscale threshold was then adjusted to convert the images into binary images.

A grid ruler was placed vertically on the soil profiles to measure the number of roots. Root diameters were measured using electronic calipers (accurate to 0.01 mm). Soil ring knives (100 cm³, diameter 50.46 mm, depth 50 mm) were used to collect soil samples from the stained and blank areas of the vertical profiles to a depth of 0–0.5 m at 0.1 m intervals (almost no traces of staining were detected at depths exceeding 0.5 m). Gene amplicon sequencing of 16S rRNA was used to determine bacterial abundance in the soil (Johnson et al., 2019). A gasoline-engine-powered soil auger with an extension rod was used to drill vertical boreholes on the adjacent slope, collecting undisturbed soil columns at depths of 0.5, 1.0, 1.5, and 2.0 m (the drill loading ring cutter had a diameter of 61.8 mm and length of 40 mm). Three replicate soil samples were collected at each depth in the nearby area. The soil samples were used to determine the shear strength index and hydraulic parameters.

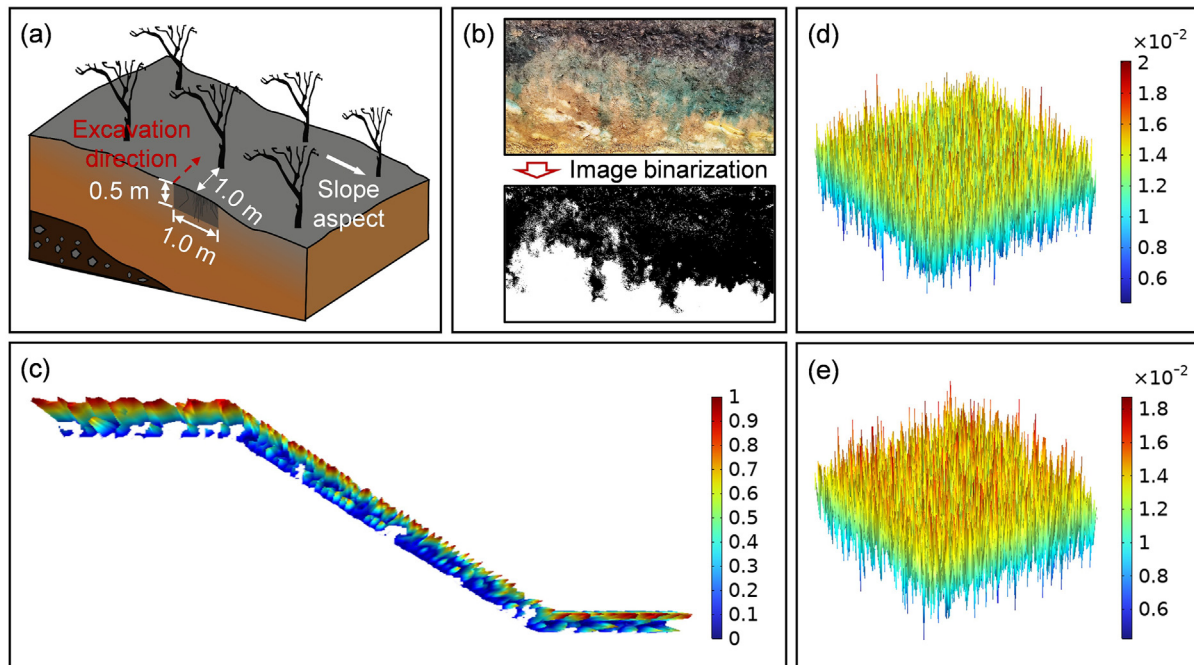


Fig. 3. Numerical modeling of soil macropore infiltration pathways: (a) location of the dye tracer experiment; (b) stained profile after chromatic aberration correction, with binary image; (c) parameterized analysis of the stained image where the red area represents macropore domain and the blue area represents matrix domain; (d) variation in soil porosity with staining depth; (e) numerical analysis of spatial distribution of soil porosity.

2.3. Saturated conductivity and soil macroporosity

The collected soil samples were saturated and left for 12 h to reach field capacity. To evaluate infiltration, we used a self-constructed constant-head apparatus. The water head height was controlled at 0.04 m, a beaker was placed on a precision electronic balance (accurate to 0.01 g) to collect the outflow, which was measured with a 5 s recording interval to reduce experimental error and to more accurately determine the time at which the flow reached stability. A data acquisition system was used to record the outflow; stability was reached when the variation in flow between three consecutive measurements did not exceed a 5% threshold. The soil pore space between the field water-holding capacity and saturated water content was used as a criterion for identifying macropores. The relationship between flow rate and pore size can be established using Poiseuille's equation, which is based on the theory of water infiltration curves under the assumption that soil pores are circular (Radulovich et al., 1989):

$$Q = \pi r^4 \Delta P / (8 \eta \tau L) \quad (1)$$

The flow rate of the steady state water flow is:

$$Q = \pi r^2 L / t \quad (2)$$

where Q is the unit flow rate (m^3/s), r is the pore radius (m), L is the length (m) of the soil column, and t is the time (s) after the initial addition of water.

The equivalent pore size of soil macropores can be derived from the joint solution of Eqs (1) and (2):

$$r = \tau L [8 \eta / (t \Delta P)]^{1/2} \quad (3)$$

where τ is the bending coefficient of the actual path of the water flow, which is generally taken as 1.2; η is the coefficient of viscosity

($\text{g} \cdot \text{cm}^{-1} \text{s}^{-1}$) of the water, and ΔP is the pressure head (m).

The maximum macropore radius can be determined from the initial watering time to the initial outflow time according to the principle of macropore-first drainage. In determining the macropore radius range, the minimum radius can be determined based on the time of stabilization of the water penetration curve, resulting in a wide range of macropore radii. The formula for calculating the interval drainage volume Q_e , based on the flow rate of the moisture outflow between any two time-points, is as follows:

$$Q_e = N A v = N \pi r^2 \tau L / t \quad (4)$$

Eq. (3) is used to classify different aperture ranges at radius intervals of 0.1–0.5 mm. The number of macropores N can then be calculated using the average value of each pore size class, r' :

$$N = Q_e t / (\pi r'^2 \tau L) \quad (5)$$

2.4. Slope stability analysis

The coupled hydromechanical framework used in this study was based on Lu et al. (2012). The Richards' equation was used to describe changes in volumetric moisture content (θ) in the soil seepage field as a function of time:

$$\frac{\partial \theta(h)}{\partial t} = \nabla \cdot K(h) \nabla H + W \quad (6)$$

where $\theta(h)$ is the soil water retention function (%), $K(h)$ is the hydraulic conductivity function (m/s), h is the pressure head (m), H is the total or hydraulic head (m), W represents sources or sinks of water. The approaches of Mualem (1976) and van Genuchten (1980) were used to describe the soil-water retention and the hydraulic conductivity function. The soil-water characteristic curve (SWCC) is

given as follows:

$$S_e = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r} = \left[\frac{1}{1 + \{\alpha|h|\}^n} \right]^m \quad (7)$$

where S_e is the effective saturation (%); θ_r and θ_s denote the residual and saturated volumetric moisture content (%), respectively; α is the inverse of the air entry value, and n is a pore-size distribution index, with $m = 1-1/n$. The hydraulic conductivity function is expressed as follows:

$$K(h) = K_s S_e^l \left[1 - \left(1 - S_e^{\frac{1}{n}} \right)^m \right]^2 \quad (8)$$

where K_s is the saturated conductivity (m/s) and l is a pore-connectivity parameter, estimated to be 0.5 in the hydraulic conductivity function of Mualem.

The Factor of Safety (FS), the most commonly used indicator for evaluating slope stability, is defined as the ratio of the shear strength (τ_F) to the gravity-driven stress (τ_D). Here, FS was defined at the node scale, because certain locations on the slope may be more prone to failure. FS is based on the displacement threshold of the unit node. The shear strength of a point on the slope in the direction of sliding is defined as follows:

$$\tau_F = c + \sigma \tan \phi \quad (9)$$

where c is the effective soil cohesion (kPa), ϕ is the effective friction angle (degrees, °), and effective stress σ is calculated as follows:

$$\sigma = \frac{\sigma_1 + \sigma_3 + 2u_w}{2} + \frac{\sigma_1 - \sigma_3}{2} \cos(2\alpha) \quad (10)$$

$$\alpha = 45^\circ + \frac{\phi}{2} \quad (11)$$

The τ_D in the sliding direction at that point is calculated as follows:

$$\tau_D = \frac{\sigma_1 - \sigma_3}{2} \sin(2\alpha) \quad (12)$$

where σ_1 and σ_3 are the major and minor principal total stresses, kPa, respectively; FS is then calculated as follows:

$$FS = \frac{\tau_F}{\tau_D} \quad (13)$$

2.5. Model implementation and parameterization

2.5.1. Mathematical models

A geometric model of the soil slope was constructed based on the topography and soil conditions of the study area, using COMSOL Multiphysics simulation software (COMSOL, Burlington, MA). The slope angle was set at 30°, height at 10 m, and length at 17 m. The area enclosed by the yellow line in Fig. 4 represents the control domain. The computational domain was extended beyond the control domain to minimize the influence of boundary conditions during slope stability assessment (Moradi et al., 2018). The slope was divided into two layers of 0–1.5 m and >1.5 m, based on differences in soil type: The top-layer comprised sandy loam disturbed by the fire, whereas the sub-layer was weakly permeable sandstone. The soil's modulus of elasticity and Poisson's ratio of the soil were assigned empirically because they only affect its elastic displacement and have a minimal effect on the stability coefficient

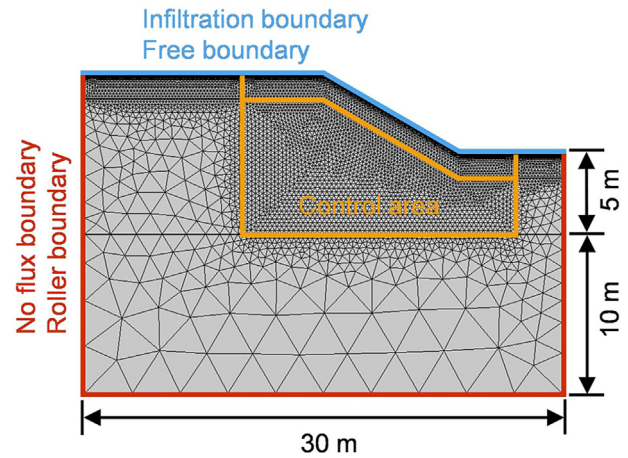


Fig. 4. Geometry, grid and boundary conditions of the slope model.

(Li et al., 2023). The soil's mechanical and hydraulic parameters assigned are shown in Table 1. Notably, the saturated conductivity (K_s) and saturated volumetric moisture content (θ_s) of the top-layer here represent the matrix domain, while the parameters of the macropore domain are represented in functional form in subsequent sections. Changes in volumetric moisture content (θ) and PWP during rainfall were obtained from the transient solution of the Darcy–Richards' equation. The boundary condition was set as the rainfall boundary at the surface. Following the fire, heavy rainfall occurred between October 2 and 5, 2022, with a maximum daily rainfall of 78.2 mm. Considering this extreme scenario, simulated rainfall was applied at 80 mm/day for 4 days. The maximum infiltration capacity of the soil is a nonlinearly variable, and it can lead to significant errors in the simulation results when the infiltration rate is directly specified, as this approach overlooks the transition of boundary conditions from a flux boundary to a pressure boundary. Therefore, a mixed boundary method was used to control the boundary conditions for infiltration and seepage (Chui & Freyberg, 2009). In our model, the left, right, and bottom boundaries of the model domain were assumed to have no displacement in the normal direction. Additionally, pressure heads of 13 m and 12 m were specified at the left and right boundaries, respectively, to simulate a constant groundwater level. The initial PWP was simulated to change linearly with elevation. The sidewall of the model was defined as a symmetrical boundary with a normal flux of 0. A monitoring point was set up at a depth of 0.5 m at the middle of the slope, at point coordinates [10.00 m, 6.61 m], to facilitate analysis of local stability. The characteristic line was placed at the $x = 10$ m on the top-layer to determine the potential failure depth in the middle of the slope, with connecting coordinates [10.00, 7.11] and [10.00, 5.11]. A very dense boundary-layer mesh was used near the ground surface to accurately simulate the transient hydrologic conditions, because the surface pressure head can change dramatically during rainfall. A coarser mesh was defined in the lower part of the slope, where the changes in surface pressure head are smaller. The complete mesh included 9971 domain cells and 5887 nodes. The geometry, computational mesh and boundary conditions are presented in Fig. 4.

2.5.2. Analysis of dye tracer images

The shallow soil layer (0–0.5 m), rich in soil macropores, was defined as heterogeneous material in the numerical model. The images obtained from dye tracing were numerically analyzed to identify the macropore and matrix domains. The image function in COMSOL allows importing images and mapping the red, green, and

Table 1
Mechanical and hydraulic parameters of slope model.

Soil layers	Dry density (kg/m ³)	Poisson's ratio (1)	Young's Modulus (MPa)	Cohesion (kPa)	Internal friction angle (°)	Saturated conductivity (m/s)	Residual Moisture (%)	Saturated moisture (%)	Parameters of VG model	
									a	n
Top-layer (0–1.5 m)	1700	0.33	20	2.7	21.0	2.2×10^{-6}	0.06	0.40	0.98	1.71
Sub-layer (>1.5 m)	1900	0.27	25	40.0	28.0	5.0×10^{-7}	0.09	0.36	0.45	1.98

blue (RGB) data of each image to scalar function output values. Each image was defined on a two-dimensional domain and is typically called using the spatial coordinates *Image* (*x*, *y*). The binary image was stitched, cropped, and rotated to perfectly match the geometric dimensions of the shallow layer of the slope model. Analytical functions were called to effectively identify physical variables within the image function. The dye tracer image was normalized to distinguish the stained regions. Intervals greater than 0.5 represented the macropore domain, whereas intervals less than 0.5 represented the matrix domain (Fig. 3(c)).

2.5.3. Parameterization of saturated volumetric moisture content and saturated conductivity

The saturated volumetric moisture content (θ_s') and saturated conductivity (k_s') of the shallow soil layer were defined as functions of macroporosity. The relationships between these parameters and macroporosity were defined based on the results of blue-dye tracing experiments. Using the normal distribution function, soil macroporosity was assigned a random distribution with staining depth within the different pore-size classes (Fig. 3(d) and (e)). The means and standard deviations of the normally distributed soil macroporosity were determined from the calculated microporosities of the stained and blank areas in each soil layer. The saturated volumetric moisture content (θ_s') of each soil layer (from 0 to 0.5 m) was calculated as the sum of macroporosity and matrix porosity for each pore-size class (Allaire et al., 2009). The saturated volumetric moisture content of the soil matrix (θ_{s0}) measured using the drying method, was 40%. Saturated water content was calculated as follows:

$$\theta_s' = [Image(x, y) > 0.5] \cdot \sum_i^m M_i(x, y) + [Image(x, y) < 0.5] \cdot \theta_{s0} \quad (14)$$

where *Image* (*x*, *y*) is the analytic function of the image, *x* and *y* are the variable coordinate ranges; M_i is the analytic function of spatial variation of soil macroporosity with staining depth for different pore-sizes classes; and *m* represents different pore-size classes. A functional relationship between the pore diameter of macropores and saturated conductivity was established, based on the differences in macropore size. Using macroporosity as the independent variable and saturated hydraulic conductivity as the dependent variable, we conducted least-squares multiple nonlinear regression for each of the four pore-size classes at each post-fire time-point and soil-layer depth. The functional expression for the saturated conductivity (k_s') in the shallow soil layer, considering macropore distribution, is calculated as follows:

$$k_s' = [Image(x, y) > 0.5] \cdot k_{s1} + [Image(x, y) < 0.5] \cdot k_{s0} \quad (15)$$

where k_{s1} and k_{s0} are the saturated conductivities of the stained and unstained areas (m/s), respectively.

3. Results

3.1. Bacterial abundance and root distribution

3.1.1. Relative abundance of soil bacterial phyla

Ten bacterial phyla were detected in the roots, excluding groups with relative abundance less than 1% (Fig. 5). The bacterial abundance fluctuated significantly after the fire. Post-fire, most of the variability in microbial abundance occurred in the shallow layer (at 0–0.3 m). At one week post-fire (T1), *Actinobacteriota*, *Proteobacteria*, and *Acidobacteriota* were the dominant bacterial groups, accounting for 18.4%, 34.1%, and 21.8%, respectively, of the sequences. At 6 months post-fire (T2), the relative abundance of *Actinobacteriota* had increased to 44.0%, whereas that of the *Proteobacteria*, and *Acidobacteriota* had decreased to 20.8% and 10.7%, respectively. At 12 months post-fire (T3), the relative abundance of *Actinobacteriota* had decreased to 29.8%, but remained higher than at T1. At T3, the relative abundances of *Proteobacteria* and *Acidobacteriota* were significantly higher than at T2, recovering 74.3% and 88.0%, respectively, of their T1 relative abundances.

3.1.2. Root distribution characteristics

The variation in root numbers was calculated for each diameter class, soil depth, and post-fire time-point (Fig. 6). Overall, the number of roots decreased gradually with increasing soil depth, and varied significantly between the post-fire time-points (Table 2). Fire affected root number most substantially in the shallow 0–0.3 m layer, and the number of fine roots (0.5–2 mm diameter) varied significantly (Fig. 6(a)). The number of fine roots (0.5–1 mm diameter) first decreased and then increased post-fire, and these roots were less abundant at T2 than at T1, by 41.0%, 33.1%, and 24.9%, respectively (Fig. 6(b)). Root number did not vary significantly between T3 and T1, indicating that new fine roots (0.5–1 mm) had already grown. The number of fine roots (1–2 mm diameter) decreased from T1 to T3, and these roots were less abundant at T2 than at T1, by 27.5%, 23.6%, and 20.8%, respectively, declining further to T3, by 35.0%, 42.2%, and 37.5%, respectively. Coarse roots (>2 mm diameter) were less abundant, by 6.8% and 9.9%, respectively, at T2 and T3 than at T1, indicating that coarse roots had undergone almost no decomposition by T3.

3.1.3. Correlations between bacterial phylum relative abundance and root number

The linear relationships between the changes in fine root numbers and in the relative abundances of the dominant bacterial phyla across the three post-fire stages were examined (Fig. 7). The change in the number of fine roots (0.5–1 mm diameter) was significantly linearly correlated with the change in the relative abundance of the dominant bacterial phyla ($R^2 > 0.80$). The change in the number of fine roots (0.5–1 mm diameter) was negatively linearly correlated with the change in the abundance of *Actinobacteriota* (Fig. 7(a)), suggesting that *Actinobacteriota* promoted the decomposition of fine roots during the first year after the fire. The change in the number of fine roots was positively correlated with changes in the abundance of *Proteobacteria* and *Acidobacteriota*

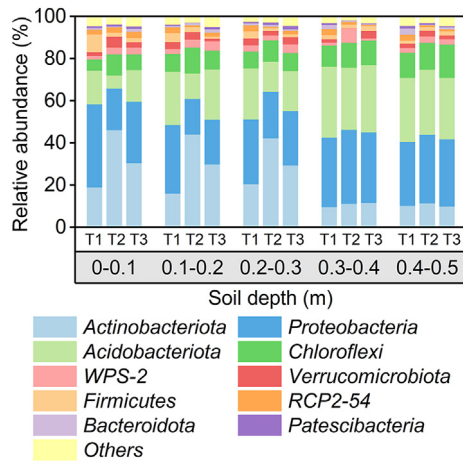


Fig. 5. Dynamics of relative abundance of soil bacterial phyla (T1, T2, and T3: one week, 6 months post-fire, and 12 months post-fire).

(Fig. 7(b) and (c)), suggesting that bacteria in these phyla are related to the recovery of fine root growth. However, the number of fine roots (1–2 mm diameter) was not correlated with the relative abundances of the dominant bacterial phyla ($R^2 < 0.50$; Fig. 7(d)–(f)), because while *Actinobacteriota* was more abundant at

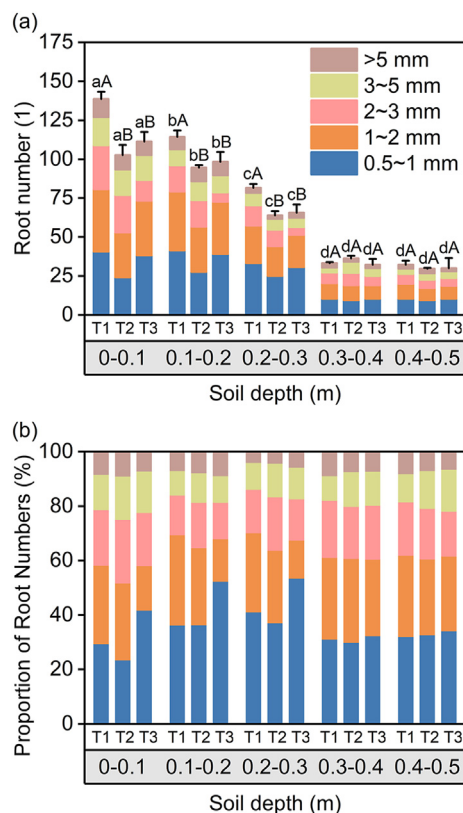


Fig. 6. Numbers and proportions of roots of different diameter classes within the cross-sectional area of the soil profile at one week, 6 months, and 12 months post-fire: (a) variation in the number of roots of different diameter classes with soil depth (means ($\pm$ SD) of 3 replicates), different lowercase letters indicate significant differences in total root numbers at different soil depths after the same fire interval ($p < 0.05$), while different uppercase letters indicate significant differences in total root numbers at the same soil depth for different time intervals ($p < 0.05$); (b) variation in the proportion of roots of different diameter classes with soil depth; the legend shows root diameter classes as 0.5–1, 1–2, 2–3, 3–5, and >5 mm.

T3 than at T1, *Proteobacteria* and *Acidobacteriota* had not yet fully recovered in abundance by T3. Roots with diameters of 1–2 mm were still decomposing at T3, and the newly formed roots had not yet reached that diameter.

3.2. Soil infiltration characteristics

The water-penetration curves and saturated conductivities at each time-point were examined (Figs. 8 and 9). Saturated conductivity was a function of the water outflow rate, and they exhibited similar trends over time. Overall, the water outflow rate and saturated conductivity decreased with increasing soil depth. The stable outflow rate was significantly greater, and was reached faster, in the stained than in blank area (Fig. 8). Consistent with this, the stained area exhibited significantly greater saturated conductivity than the unstained area (Fig. 9). The fire altered the flow rate and permeability primarily in the shallow soil layer (0–0.3 m), consistent with its effects on root distribution. In the shallow soil layer, the stained area exhibited 1.85 and 2.64 times greater average saturated conductivity at time-points T2 and T3, respectively, than at T1. In contrast, the saturated conductivity of the unstained area remained almost unchanged.

3.3. Soil macroporosity

3.3.1. Characterization of macroporosity distribution

Based on the definition of the equivalent pore diameter for soil macropores (Lamandé et al., 2011), we divided soil macroporosity into four classes, based on minimum and maximum values of the classes: 0.5–1.0, 1.0–1.5, 1.5–2.0, and 2.0–2.5 mm. The macroporosity (%) for the different radius ranges is shown in Fig. 10. Overall, the stained area exhibited greater macroporosity than the unstained area. The macroporosity decreased gradually with soil depth. Post-fire, macroporosity varied over time primarily in the 0–0.3 m soil layer, consistent with the changes in root distribution and saturated conductivity, increasing over time. The average macroporosities were 1.51 and 2.12 times greater, respectively, at T2 and T3 than at T1.

3.3.2. Saturated conductivity and macroporosity

The functional relationship between saturated conductivity and soil macroporosity is expressed as Eq. (16). The range of saturated conductivity k_{s1} predicted by the fitted equation was consistent with our experimental results (Fig. 9), but exhibited a better fit ($R^2 = 0.985$) (Fig. 11(a)). The parameterized results for saturated moisture content (Eq. (14)) are presented in Fig. 11(b). The spatial variation in saturated conductivity (k_{s1}) with soil depth and macropore distribution can be obtained by substituting Eq. (16) into Eq. (15) (Fig. 11(c)).

$$k_{s1} = 1.02 \times 10^{-7} + 4.41 \times 10^{-5}\theta_1 + 7.35 \times 10^{-5}\theta_2 + 1.03 \times 10^{-4}\theta_3 + 4.31 \times 10^{-5}\theta_4 + 2.22 \times 10^{-6}\theta_1^2 + 3.70 \times 10^{-6}\theta_1\theta_2 + 5.18 \times 10^{-6}\theta_1\theta_3 + 2.17 \times 10^{-6}\theta_1\theta_4 + 6.16 \times 10^{-6}\theta_2^2 + 8.63 \times 10^{-6}\theta_2\theta_3 + 3.61 \times 10^{-6}\theta_2\theta_4 + 1.21 \times 10^{-5}\theta_3^2 + 5.06 \times 10^{-6}\theta_3\theta_4 + 2.12 \times 10^{-6}\theta_4^2 \quad (16)$$

where θ_1 to θ_4 represent the soil macroporosity in the pore diameter classes 0.5–1.0, 1.0–1.5, 1.5–2.0, and 2.0–2.5 mm, respectively.

Table 2
Results of the two-way ANOVA showing the *p*-values for root number, relative abundance, and saturated conductivity of the stained area.

	Soil depth (D)	df (D)	Post-fire time (T)	df (T)	D × T	df (D × T)
Root number (1)	<0.05	4	<0.05	2	<0.05	8
Relative abundance (%)	0.994	4	<0.05	2	0.999	8
Saturated conductivity of stained area (m/s)	<0.05	4	<0.05	2	<0.05	8

Note: The “df” refers to the degrees of freedom for each variable.

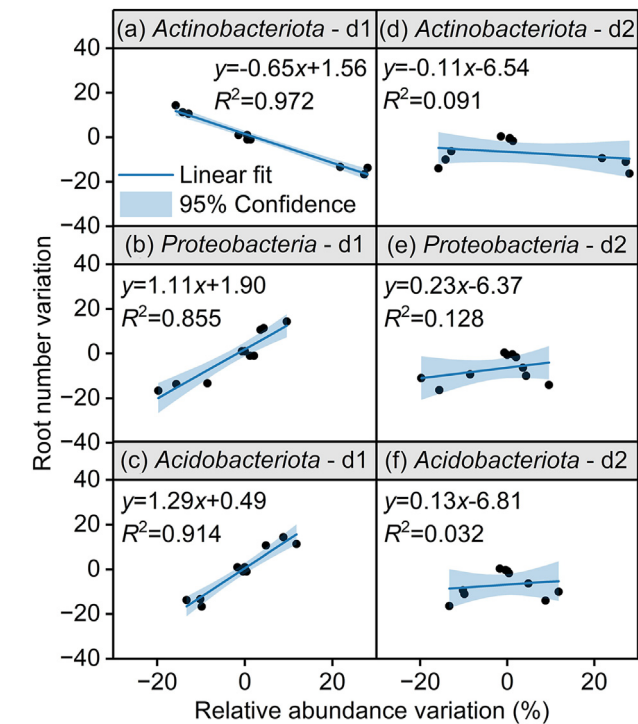


Fig. 7. Correlations between changes of relative abundance of the dominant bacterial phyla (including *Actinobacteriota*, *Proteobacteria*, and *Acidobacteriota*) and changes in the number of fine roots, d1 and d2 represent the fine root systems with diameters of 0.5–1 mm and 1–2 mm, respectively, and the blue shaded areas represent 95% confidence intervals.

3.4. Slope seepage field and stability

3.4.1. Dynamics of slope seepage field

Cloud plots of the *PWP* of the slope during the simulated four days of heavy rainfall at each post-fire time-point, and variation in *PWP* with soil depth along the characteristic profile lines, were examined (Fig. 12). The changes in the seepage field can be divided into three stages. First, the surface soil became wet; the initial *PWP*, which was linearly distributed with depth, increased significantly over the rainfall period at the surface of the slope. Although the wetting front moved rapidly downward, *PWP* did not change in deeper soil layers during the rainfall period. Under the influence of continuous rainfall, water accumulated first at the toe of the slope, accompanied by downslope seepage when the wetting front reached the structural surface of the soil layer. Water spread from the downslope to the upslope positions, and the *PWP* of the upper soil layer gradually increased. The accumulated water at the toe of the slope gradually permeated through the structural surface into the lower soil layers, although the groundwater level did not increase notably. Comparing the time-points, T1 exhibited the highest shallow-layer *PWP* during the simulated rainfall period. The slope exhibited higher macroporosity and better soil permeability at T2 and T3 than at T1, allowing the water to infiltrate rapidly into

the deeper soil layers, leading to higher *PWP* in the deeper layers at T2 and T3. At T3, the greater macroporosity of the soil facilitated faster propagation of pressure. At all three time-points, the shallow layers approached saturation during rainfall. Slope surface *PWP* gradually increased and approached 0 kPa, with the *PWP* distribution curves tending to converge.

3.4.2. Dynamics of slope stability

Cloud plots of slope stability and the variation in the Factor of Safety (*FS*) of the characteristic lines with soil depth of the slope at each post-fire time-point during the simulated four days of heavy rainfall were examined (Fig. 13). During the heavy rainfall, unstable areas developed first in the upper and middle sections of the slope, then expanded into the deeper soil layers, with the water infiltrating deeper into the soil. Although the toe of the slope exhibited a high *PWP* (Fig. 12), it remained stable because water drained to the right of the slope. The distribution of soil macropores varied between the post-fire time-points, leading to differences in the soil water storage capacity of the shallow soil layer. At the end of the rainfall event, the potential failure depths (*Z_f*, corresponding to the critical failure depth at *FS* = 1) in the middle of the slope were 0.36 m, 0.25 m, and 0 m, respectively, at T1, T2, and T3. At T3, the failure depths were shallow (*Z_f* < 0.5 m), and the slope had not yet failed. At T3, the *FS* at the monitoring point decreased within the first 1.8 d of rainfall (Fig. 14), owing to the greater macroporosity at this time-point, which allowed the water pressure to propagate more rapidly into the 0.5 m soil layer. On the first day of rainfall, T2 and T3 exhibited lower *FS* than T1, by 3.55% and 8.68%, respectively. However, the water pressure propagated rapidly into the deeper layers, causing *FS* at the monitoring point to be higher after 1.8 days of rainfall. At the end of the rainfall event, T2 and T3 exhibited greater *FS* than T1, by 1.87% and 2.81%, respectively.

4. Discussion

4.1. The influence of microbial activity on root decay

Root activity is influenced by several factors, including soil microorganism diversity, soil temperature, humidity, pH, and nutrient levels, all of which can affect root growth and decay (Zhao et al., 2024). The soil temperature required for normal root growth and respiration typically ranges from 12 to 35 °C, with soil moisture ranging from 12.2% to 30.1% (Gregory, 2006) and pH ranging from 3.5 to 4.5 (Li et al., 2019). The post-fire surface temperature, humidity, and pH were all appropriate for the normal growth of *Pinus massoniana* (Table A1), and soil nutrient availability was sufficient to meet its root growth requirements during both the growing and non-growing seasons (Fig. S1) (Jian et al., 2021). Therefore, we conclude that root decomposition at the study site was influenced primarily by bacterial community activity, rather than by other factors. *Actinobacteriota* are saprophytic bacteria that promote plant root decomposition by producing cellulases and ligninases (Daquioag & Penuliar, 2021; Fleishman et al., 2023). *Proteobacteria* and *Acidobacteriota* promote root growth by regulating plant metabolic activity (Cao et al., 2024). In this study, post-fire changes in the abundance of these dominant phyla were significantly

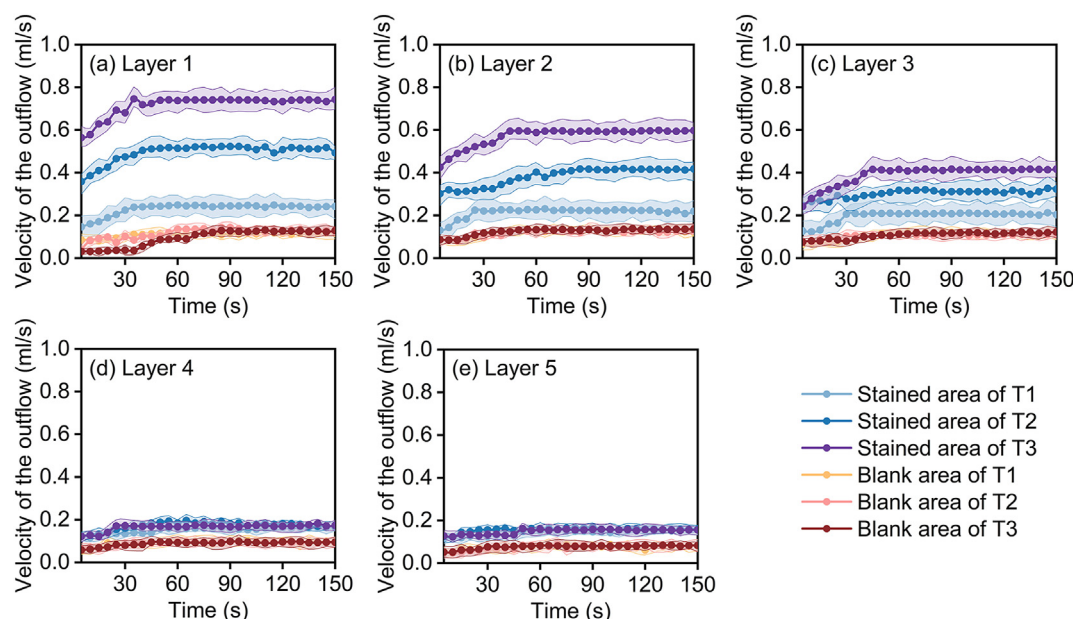


Fig. 8. Water infiltration curves, (a–e) soil depths of 0–0.1, 0.1–0.2, 0.2–0.3, 0.3–0.4, 0.4–0.5 m, respectively.

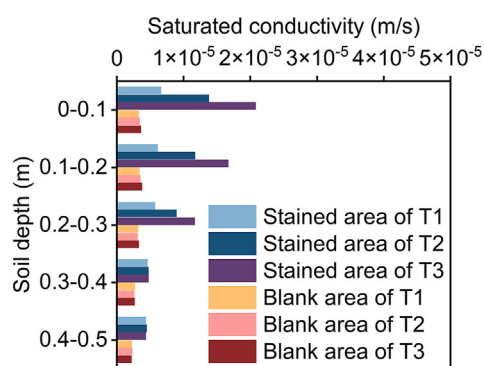


Fig. 9. Saturated conductivity (K_s) at different soil depths (average of 3 replications).

linearly correlated with changes in the number of fine (0.5–1 mm diameter) roots (Fig. 7), but, surprisingly, not with changes in the number of coarse (1–2 mm diameter) roots. This difference may be related to the different specific surface areas of the fine and coarse roots. After fire disturbance, smaller roots are more easily decomposed by soil microorganisms, whereas larger roots require longer decomposition times (Vergani et al., 2017).

4.2. The influence of post-fire root decay on soil macropores

Plant root size, soil macroporosity, and saturated conductivity are quantitatively related (Gackiewicz et al., 2022; Shao et al., 2020; Wei et al., 2019). The saturated conductivity in burnt woodland was approximately half of that in unburnt woodland in the same study

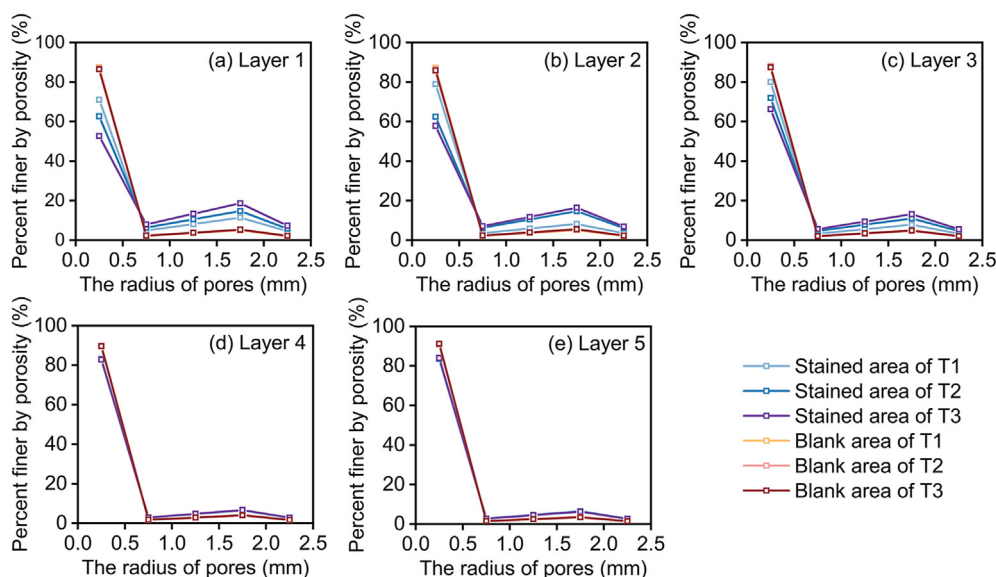


Fig. 10. Macroporosity (as proportion) by pore diameter class at each soil depth.

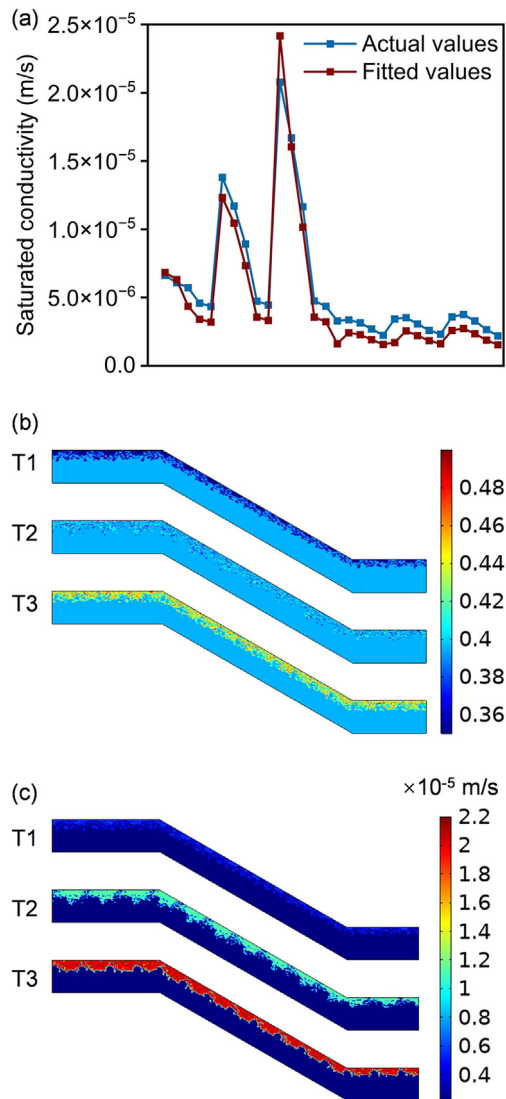


Fig. 11. Predicted saturated conductivity by soil macropore volume: (a) measured and fitted saturated conductivity; (b) cloud plot of saturated moisture content; (c) cloud plot of saturated conductivity fitted via multiple nonlinear regression.

area at T1 (Wang et al., 2020). This might be attributed to the formation of hydrophobic substances from root combustion, and to ash blocking the pores of the soil structure (Lei et al., 2022; Mao et al., 2019). In contrast, the water outflow rate and saturated conductivity in the stained area of the shallow soil layer was significantly greater (by 1.85 times) at T2 than at T1 (Figs. 8 and 9). This is due both to the disappearance over time of hydrophobic compounds and ash in the soil and to the decomposition of 0.5–1 mm fine roots and the resulting increase in macroporosity (which was 1.51 times greater at T2 than at T1) (Fig. 10). In the stained area, the shallow soil-layer saturated conductivity and macroporosity were 2.64 and 2.12 times greater at T3 than at T1, respectively (Figs. 9 and 10). Although low-density fine roots may fill old root channels (Zhang et al., 2023), the decomposition of the coarse 1–2 mm diameter roots releases organic matter that promotes soil particle aggregation and the formation of root channels, causing macroporosity to increase. According to the unsaturated flow theory, soil moisture absorption involves a reduction in matric suction and acceleration of unsaturated conductivity. Soils with more macropores tend to have lower air-entry pressure values

(Antinoro et al., 2017). Water flow was proportional to the fourth power of pore diameter during vertical water infiltration (Eq. (1)), suggesting that the additional macropores created by root decomposition accelerate water infiltration. Post-fire changes in soil structure have been described using electron microscopy, and channel formation due to the continuous decomposition of dead roots increases soil saturated conductivity (Lei et al., 2022). Soil porosity quantitatively influences saturated conductivity (Zhang et al., 2023). In this study, multivariate nonlinear regression was used to establish the functional relationship between saturated conductivity and the frequency of soil macropores of different diameter ranges ($R^2 = 0.985$). Microstructural observations of soil by Gackiewicz et al. (2022) confirmed that the multiple regression model can accurately describe the relationship between them. The frequency of coarse roots (>2 mm diameter) did not change significantly during the post-fire study period. However, Zhang et al. (2023) found that 0.5–2 mm diameter roots are more important for soil hydraulic conductivity than finer roots. Future research is required to quantitatively examine the impact of root decay and recovery processes on soil hydraulic conductivity and to identify the equilibrium point and time-point at which changes in root distribution result in the highest conductivity.

4.3. Influence of soil macropores on slope seepage field and stability

The changes in slope seepage fields under the intense rainfall conditions at different post-fire time-points were analyzed from the perspective of unsaturated seepage. The deeper mudstone layers exhibited a lower FS than the shallower layers, and the slope was relatively stable, owing to the greater friction angle and lower permeability of the deeper soil. The infiltration of extreme rainfall gradually reduced the matric suction of the surface sandy loam layer, and the PWP of the slope surface gradually increased, approaching 0 kPa (Fig. 12). The danger of slope failure increased from the deeper to the shallower layers (Fig. 13), consistent with the triggering of shallow landslides during rainfall events (Moradi et al., 2018; Li et al., 2023). Rainfall intensity and duration are the key factors affecting the stability of rainfall-affected slopes (Fan et al., 2020). However, for rainfall of short-duration and low-intensity, infiltration into the slope is insufficient, resulting in minimal changes in PWP, and no significant differences in the deformation of both homogeneous and heterogeneous slopes (Shao et al., 2015). Therefore, we considered actual rainfall scenarios within the study area (Fig. 2), setting extreme conditions (80 mm/d $\times$ 4 d) as the rainfall boundary of the numerical model, allowing for a more direct observation of the impact of macropores on the slope infiltration process.

Soil macropores may affect slope stability through two mechanisms, by enhancing the drainage capacity of the soil mass and by inducing rapid pressure build-up during heavy rainfall (Sidle & Bogaard, 2016; Tao et al., 2020). In this study, the shallow soil layer exhibited higher PWP at T1 than at T2 and T3 during the rainfall events (Fig. 12). This is because the slope was relatively unsaturated at T1, making it easier to reach a near-saturated state. The lower frequency of macropores caused PWP to accumulate more rapidly in the shallow soil layers (Tao et al., 2020). In contrast, the saturated volumetric moisture content of the slope was greater at T3, and thus did not become fully saturated during the same rainfall event. The presence of macropores allowed the water pressure to propagate more quickly to the deeper layers (Yeh et al., 2021). Some studies have shown that an increase in macropores or cracks in the soil can accelerate slope instability (Kukemilks et al., 2018; Abdollahi et al., 2023). However, we observed that under extreme rainfall, the contribution of macropores to the local stability of the slope decreased, at first rapidly and later more slowly

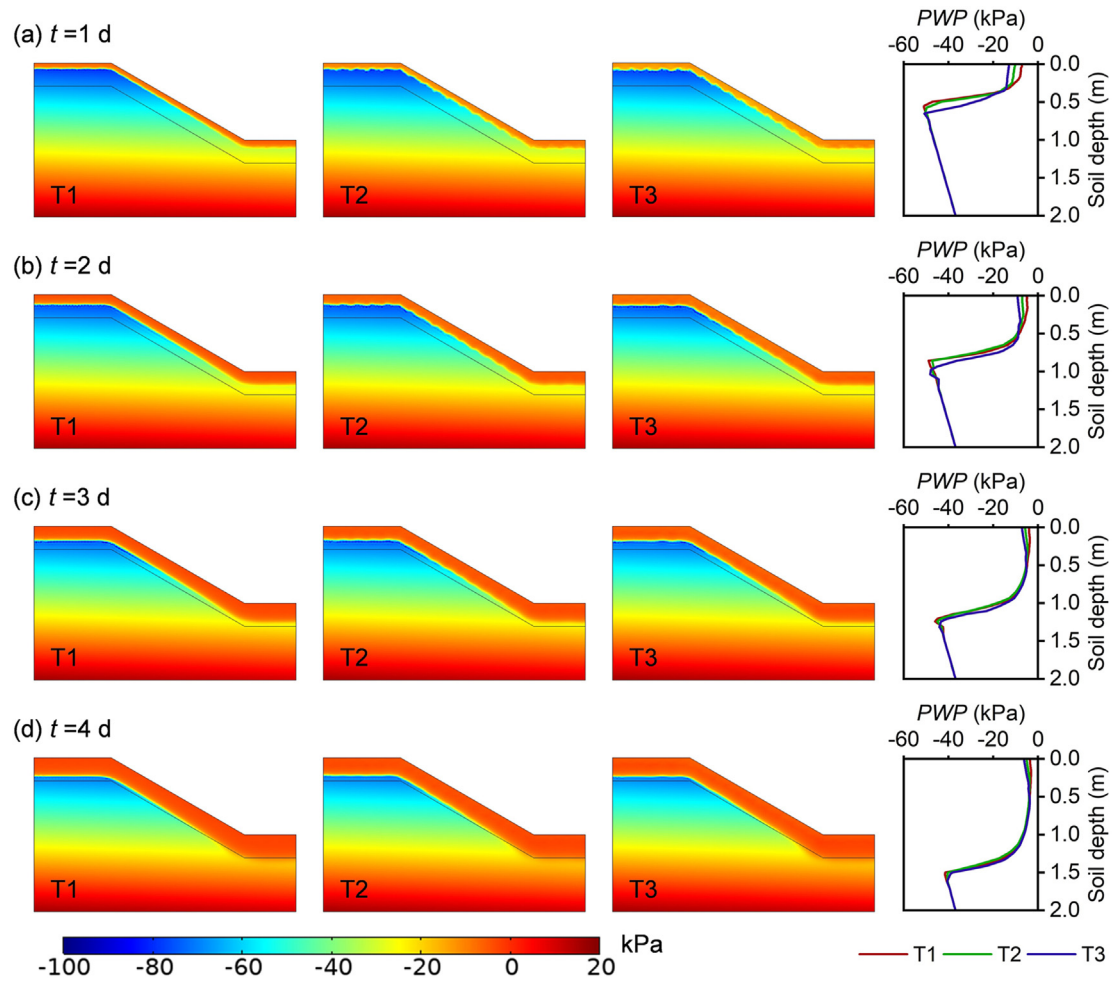


Fig. 12. Cloud plots of the simulated changes in pore water pressure (PWP) over the four days of rainfall, line charts (right) show the PWP along characteristic lines in the middle of the slope during rainfall at each of the three post-fire time-points.

(Fig. 14). The greater macroporosity affected slope stability more in the early stages of rainfall. In contrast, from day 1.8 of rainfall until the end of the event, FS was slightly greater at T3 than that at T1, because in soils with greater macroporosity, the water flows more strongly in the lateral direction along the bedrock surface (Moradi et al., 2018). In our simulation, the water drained from the right side of the slope, and PWP was lower at the monitoring point location (in the middle of the slope), thus reducing the FS of the slope. In real-world scenarios, the slopes may be steeper and the soils shallower, with lower shear strength and poor drainage, leading to slope instability during the pre-rainfall period.

4.4. Limitations and improvements

Under the influence of climate change and human activities (e.g., wildfires or deforestation), plant root decay can severely impact the soil structure (Abdollahi et al., 2023). Highly heterogeneous slopes pose a challenge for landslide early warning systems (Liu et al., 2021). Although significant advancements have been made to hydraulic models for quantifying the stability of heterogeneous slopes (Shao et al., 2018; Yeh et al., 2021), capturing the seepage characteristics of highly heterogeneous slopes remains challenging. This study explores and theoretically predicts the hydraulic behavior of slopes influenced by macropores. In traditional dual permeability model simulations of heterogeneous slopes, the

moisture content showed a linear variation trend during rainfall (Shao et al., 2015). However, the method of this study integrated the spatial distribution characteristics of soil macropores with the coupled analysis of slope hydromechanical behavior (Figs. 10 and 11). The wetting front displayed a zigzag downward movement, and PWP exhibited a nonlinear increase with macropore distribution (Fig. 12), thereby more accurately reflecting the infiltration-enhancing effect of macropore channels (Sidle & Bogaard, 2016). Fig. 15 illustrated the stability changes at a depth of 0.5 m in heterogeneous slopes (using the post-fire 12-month slope model as an example) compared to those in homogeneous slopes during rainfall. In the early rainfall stage, macropore effects contribute up to 9.8% to the FS, whereas at the end of rainfall, the contribution decreases to 3.9%. This is significantly greater than the results calculated using the dual permeability model (i.e., 0.5–2.5%) (Shao et al., 2018), demonstrating the superior performance of the new method we proposed for analyzing the stability of heterogeneous slopes. For highly heterogeneous soil slopes (e.g., fire-affected soils in regions such as the Mediterranean coast (Castellet-Viciano et al., 2024), western North America (Abdollahi et al., 2023), and south-western China), considering the spatiotemporal variability of macropores can improve the accuracy of slope stability analysis. Moreover, differences may arise when the proposed model is applied to other soil types (e.g., sandy and clay soils). Sandy soils have greater permeability, and the impact of macropores becomes

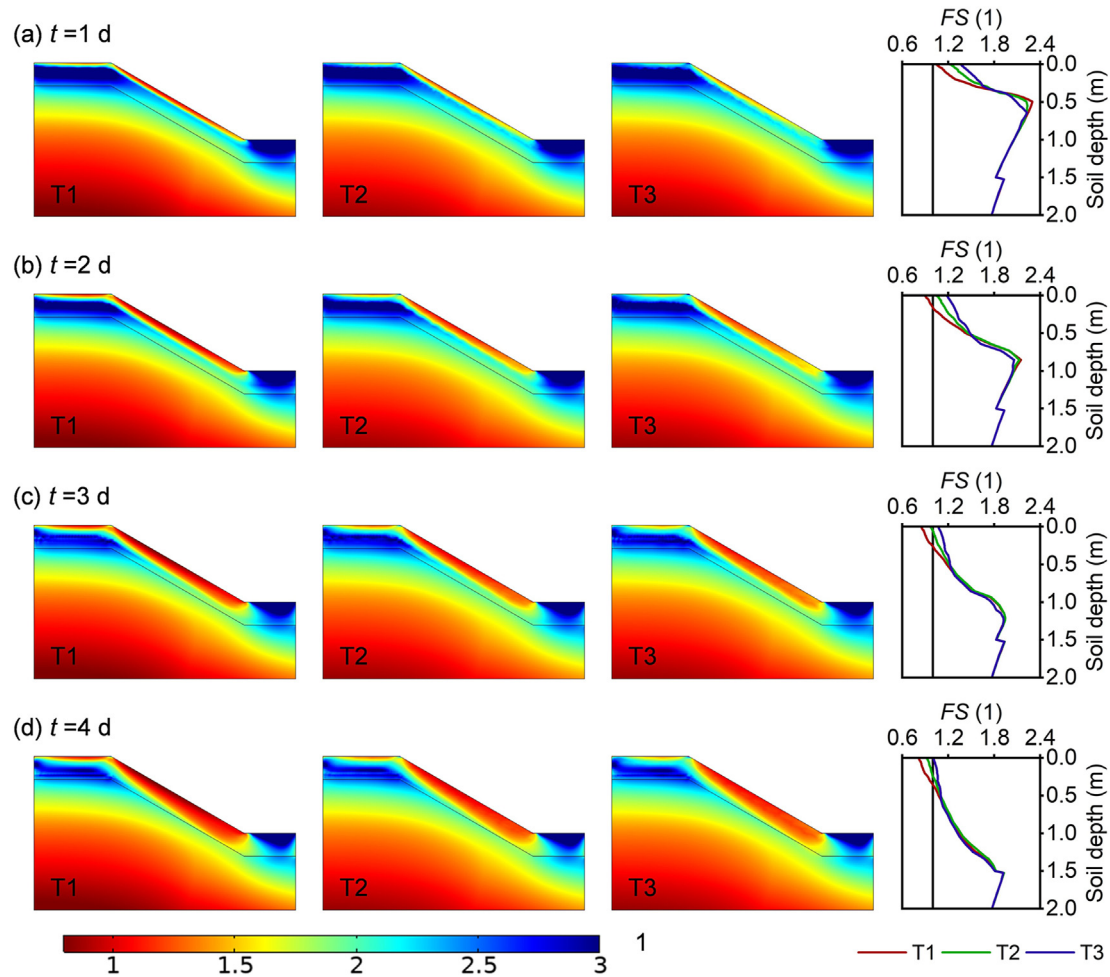


Fig. 13. Cloud plots of Factor of Safety (FS) over the four days during rainfall, line charts (right) show the FS along characteristic lines in the middle of the slope during rainfall at each of the three post-fire time-points.

apparent under extreme rainfall conditions, where moisture rapidly spreads to deeper layers through preferential flow paths (Abdollahi et al., 2023). Conversely, clay matrix domains have lower permeability, with macropores serving as the primary pathways for moisture infiltration. The fissure characteristics of clay can lead to lateral diffusion of moisture along the fissures, resulting in localized accumulation of PWP (Sidle & Bogaard, 2016). These findings indicate that image analysis methods could capture more

accurately the hydromechanical behavior of slopes. Notably, the functional relationship between saturated conductivity and soil macroporosity requires further validation before it can be adapted to other regions with varying soil or vegetation conditions. Furthermore, under extreme disturbances, microbial activity decomposing cellulose, hemicellulose, and lignin in root tissues also reduces the tensile strength of fine roots over time (Zhu et al., 2020), affecting the long-term prediction of slope stability (Sidle & Bogaard, 2016). Future research should consider the contribution of root adhesion strength changes to slope stability to enhance the accuracy of numerical slope simulations.

4.5. Implications for management

The heterogeneous slope analysis method proposed in this study provides valuable insights regarding early warning of shallow landslides, particularly in forested areas affected by logging or wildfires. Additionally, macropores can accelerate water flow in shallow soil layers, leading to reduced water pressure. Establishing drainage systems in regions with abundant rainfall can facilitate the discharge of accumulated water from slopes and provide valuable insights regarding soil and water conservation engineering in mountainous areas.

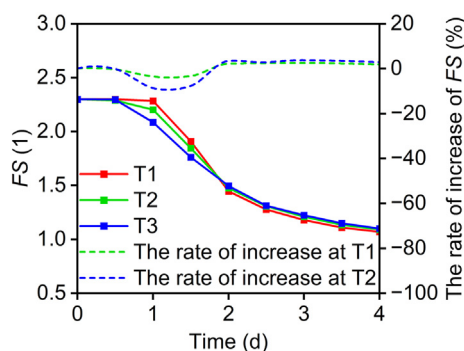


Fig. 14. Factor of Safety (FS) at each of the monitoring-point positions during rainfall at the three post-fire time-points.

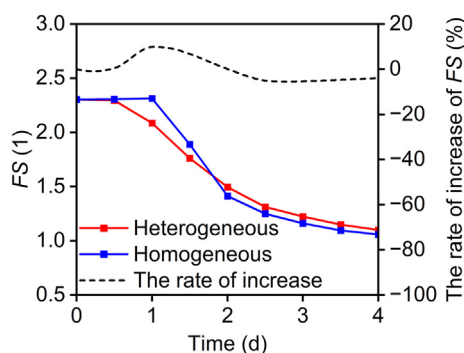


Fig. 15. Comparison of slope stability between heterogeneous and homogeneous slopes during rainfall (the black dashed line represents the contribution rate of macropores to Factor of Safety (FS), calculated using the formula: $S = (FS_{\text{Homogeneous}} - FS_{\text{Heterogeneous}}) / FS_{\text{Homogeneous}}$).

5. Conclusions

This study examined changes in the relative abundance of bacterial phyla, the spatial distribution of plant roots, and the distribution of soil macropores in forest soil one week, 6 months, and 12 months post-fire. The shallow slopes responded most to the wildfire disturbance. The activity of the dominant bacterial phyla (*Actinobacteriota*, *Proteobacteria*, and *Acidobacteriota*) led to root decay, increasing macropore formation and thus accelerating slope instability. Shallow soil layer macroporosity was 1.51 and 2.12 times greater, respectively, at 6 and 12 months than at one week post-fire. The additional macropores generated by root decay accelerate water infiltration, causing the saturated conductivity of the soil to be 1.85 and 2.64 times greater, respectively, at 6 and 12 months than at one week post-fire. In the simulation, under extreme rainfall (80 mm/d $\times$ 4 d), the presence of macropores facilitated infiltration into the deeper layers. During rainfall, slope stability was governed primarily by the distribution of PWP. The role of macropores in accelerating slope instability was most pronounced within the first 1.8 d of rainfall, and particularly on the first day of rainfall. The slope's FS was 3.55% and 8.68% lower, respectively, at 6 and 12 months than at one week post-fire. However, preferential drainage via macropores caused slope stability during the end of the simulated rainfall event to be slightly greater at 6 and 12 months than at one week post-fire. These findings highlight the importance of soil macropores generated by root decay after a wildfire in triggering shallow landslides, and elucidate the effects of the spatial distribution of macropores on slope failure mechanisms. The proposed slope stability analysis framework, coupled with analysis of macroporous soil seepage paths, may provide an accurate representation of infiltration into heterogeneous slopes, along with a reference for slope hydrology research and for the development of shallow-landslide early warning systems.

CRedit authorship contribution statement

Zihan Qi: Writing – review & editing, Writing – original draft, Investigation, Data curation, Conceptualization. **Yunqi Wang:** Supervision, Project administration, Funding acquisition. **Tong Li:** Methodology. **Xiangjun Yan:** Investigation. **Yue Lan:** Investigation. **Xiaoming Zhang:** Resources. **Peng Li:** Resources. **Liqun Lyu:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing

financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.iswcr.2025.02.007>.

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